

4	(a)	(i)	$pV = NkT$ $(2.5 \times 10^5) \times (3.8 \times 10^{-2}) = N \times (1.38 \times 10^{-23}) \times (181 + 273)$ $N = 1.5 \times 10^{24}$	1 1
		(ii)	From first law of thermodynamics, increase in internal energy = heat energy supplied + work done on the system Since the process took place at constant volume, work done = 0 Thus, increase in internal energy = 2700 + 0 = 2700 J (Since it is an increase in the internal energy, the change in internal energy is positive 2700 J)	1 1
	(b)		$\Delta E_K = 3/2 N k (\Delta T)$ $2700 = 3/2 (1.5 \times 10^{24}) (1.38 \times 10^{-23}) \times \Delta T$ $\Delta T = 87$ $T = 181 + 273 + 87 = 541 \text{ K}$	1 1
5	(a)	(i)	Given $I = nAve$	